

A finiteness result for $y^3 + y = x^4 + x$

A correction to the proposed infinitude assertion

Status of the answer. The assertion of infinitely many integer solutions is false. The proof below uses Siegel's theorem and the genus formula for a smooth plane curve; their statements are supplied, but they are not proved. Thus the argument is *unconditional*, but it does *not* meet the requested fully self-contained standard. No complete enumeration of the integer solutions is claimed. No internet search was used.

Theorem 1. *There are only finitely many pairs $(x, y) \in \mathbb{Z}^2$ satisfying*

$$y^3 + y = x^4 + x. \quad (1)$$

Proof. We verify explicitly the geometric hypotheses of the finiteness theorem. Homogenizing (1) gives the projective plane curve

$$C : F(X, Y, Z) = Y^3Z + YZ^3 - X^4 - XZ^3 = 0.$$

All smoothness and irreducibility checks below are over $\mathbb{C}$.

1. The point at infinity is unique and smooth. Setting $Z = 0$ gives $X = 0$, so the only point at infinity is $P = [0 : 1 : 0]$. Since

$$F_Z = Y^3 + 3YZ^2 - 3XZ^2, \quad F_Z(P) = 1,$$

the gradient is nonzero at P .

2. There are no affine singularities. For $f(x, y) = y^3 + y - x^4 - x$, a singular point would satisfy

$$f = 0, \quad f_x = -4x^3 - 1 = 0, \quad f_y = 3y^2 + 1 = 0.$$

Consequently $x^3 = -1/4$ and $y^2 = -1/3$. Substitution into $f = 0$ then gives

$$\frac{2}{3}y = \frac{3}{4}x, \quad y = \frac{9}{8}x, \quad x^2 = -\frac{64}{243}.$$

But these equations also force

$$x = \frac{x^3}{x^2} = \frac{243}{256},$$

whose square is positive, contradicting $x^2 = -64/243$. Thus C is smooth everywhere.

3. The curve is geometrically irreducible. Suppose $F = GH$, with nonconstant homogeneous complex polynomials of degrees $d, e > 0$. Since $F(X, Y, 0) = -X^4$, their restrictions to $Z = 0$ must be cX^d and $c'X^e$, where $cc' = -1$. Hence both factors vanish at P , giving $\nabla F(P) = H(P)\nabla G(P) + G(P)\nabla H(P) = 0$. This contradicts step 1 and proves irreducibility.

4. Apply the genus formula and Siegel's theorem. The genus formula states that a smooth geometrically irreducible projective plane curve of degree d has genus $(d-1)(d-2)/2$. Here

$$g(C) = \frac{(4-1)(4-2)}{2} = 3.$$

The form of **Siegel's theorem** needed here is: an affine geometrically irreducible curve over $\mathbb{Q}$ whose smooth projective completion has positive genus has only finitely many integer points in any fixed affine embedding. Applying it to $C \setminus \{P\}$, with coordinates (x, y) , proves the theorem. $\square$

Intuition and elementary information

The leading terms suggest trying

$$x = t^3, \quad y = t^4,$$

because then $y^3 = x^4 = t^{12}$. In the actual equation this leaves $t^4 = t^3$, so the trial works only for $t = 0, 1$. The lower-order terms therefore matter even though they are small compared with the leading terms. This failed trial alone would not rule out some other infinite family.

The geometric calculation resolves that difficulty. The simpler curve $y^3 = x^4$ has a singularity at the origin and admits the displayed parametrization. For the curve in (1), the derivative calculation shows that no singularity remains, including over the complex numbers. Its smooth quartic structure gives genus 3, and Siegel's theorem supplies the arithmetic finiteness conclusion. Merely comparing the degrees 3 and 4 would not establish finiteness.

Some useful information can also be proved entirely elementarily. Put $h(u) = u^3 + u$. For real $v > u$,

$$h(v) - h(u) = (v - u)(v^2 + uv + u^2 + 1) > 0,$$

so h is strictly increasing. For every integer x , $x^4 + x \geq 0$; hence any integer solution has $y \geq 0$. At $x = -1, 0, 1$, strict monotonicity determines exactly the solutions

$$\boxed{(-1, 0), \quad (0, 0), \quad (1, 1).}$$

For $a = |x| \geq 2$,

$$x^4 + x - h(a) \geq a^4 - a - a^3 - a = a(a^2(a - 1) - 2) > 0,$$

so every further solution would satisfy $y > |x|$.

Elementary proposition. *The only integer solutions of (1) for which x is an integer cube are $(-1, 0), (0, 0), (1, 1)$.*

Proof. Write $x = t^3$. The cases $t = -1, 0, 1$ were handled above. Suppose $|t| \geq 2$, and put $q = t^4 \geq 16$. Then

$$x^4 + x = q^3 + t^3 < q^3 + q = h(q),$$

because $t^3 < q$. On the other hand, $|t|^3 \leq q$ gives

$$\begin{aligned} q^3 + t^3 - h(q - 1) &= 3q^2 - 4q + 2 + t^3 \\ &\geq 3q^2 - 5q + 2 = (3q - 2)(q - 1) > 0. \end{aligned}$$

Thus $h(q - 1) < x^4 + x < h(q)$. Strict monotonicity makes $h(y) = x^4 + x$ impossible for integral y , since no integer lies strictly between $q - 1$ and q . $\square$

The proposition settles every cube value of x , but does not settle the noncube values. The global result proved on the first page is finiteness; it supplies neither an explicit bound for $|x|$ nor a proof that the three displayed solutions are the complete list. The reliance on Siegel's theorem is the substantive gap between this argument and a fully self-contained treatment.